

Time-ordered expansion for the derivation of time-convolutionless master equations

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I. INTRODUCTION

II. PROBLEM SETTING AND KNOWN RESULT

A. Problem setting

Let us consider a quantum system defined on a Hilbert space $\mathcal{H}$, and let $\mathfrak{B}(\mathcal{H})$ denote the set of bounded operators acting on $\mathcal{H}$. Throuout the entire paper we only consider finite-dimensional systems, i.e. $\dim(\mathcal{H}) < \infty$, $\mathcal{H} \simeq \mathbb{C}^n$ and $\mathfrak{B}(\mathcal{H}) \simeq \mathbb{C}^{n \times n}$. The state of such a system is described by a density operator $\rho \in \mathfrak{D}(\mathcal{H})$ with $\mathfrak{D}(\mathcal{H}) = \{\rho \in \mathfrak{B}(\mathcal{H}) | \rho = \rho^\dagger \geq 0, \text{tr} \rho = 1\}$. We here assume that the state of the system evolves according to the Von-Neumann master equation

$$\dot{\rho}_t = \mathcal{L}(\rho_t) \quad (1)$$

where $\rho_0 \in \mathfrak{D}(\mathcal{H})$ is the system's initial condition, and $\mathcal{L}$ is the celebrated Lindblad (or GKLS) generator [1, 2]:

$$\mathcal{L}(\rho) = -i[H, \rho] + \sum_k \mathcal{D}_{L_k}(\rho) \quad (2)$$

where $H = H^\dagger$ is the system's Hamiltonian and $\mathcal{D}_L(\rho) = L\rho L^\dagger - \frac{1}{2}\{L^\dagger L, \rho\}$ represents the system's dissipation toward a Markovian environment represented by a set of operators $\{L_k\}$ called noise operators. The evolution of the quantum system is then described by the exponential map $\rho_t = e^{\mathcal{L}t}(\rho_0)$. Importantly, $e^{\mathcal{L}t}$ generates a quantum dynamical semigroup, i.e. a one-parameter semigroup of completely positive (CP) and trace preserving maps (TP). In the following we will say that a super-operator is of *Lindblad-type* if it can be put in the form of equation (2).

We next assume to be given a CPTP, not necessarily orthogonal (w.r.t. $\langle \cdot, \cdot \rangle_{HS}$), projector $\mathcal{P} : \mathfrak{B}(\mathcal{H}) \rightarrow \mathfrak{B}(\mathcal{H})$, $\mathcal{P}^2 = \mathcal{P}$, whose image we denote by $\mathcal{V} \equiv \text{Im} \mathcal{P}$. Furthermore, for the sake of simplicity, we assume that $\mathcal{P}$ admits a full rank fixed point, i.e. $\exists \rho \in \mathfrak{D}(\mathcal{H})$, $\rho > 0$ such that $\mathcal{P}(\rho) = \rho$. As we shall see in more details later (in Sec.IV A) such a projector is in fact a conditional expectation and enjoys some useful properties. For the moment however, it is sufficient to know that $\mathcal{P}$ can be decomposed in two factors $\mathcal{P} = \mathcal{J}\mathcal{R}$ which take the name of reduction and injection maps $\mathcal{R} : \mathfrak{B}(\mathcal{H}) \rightarrow \mathcal{A}$ and $\mathcal{J} : \mathcal{A} \rightarrow \mathfrak{B}(\mathcal{H})$ such that $\mathcal{R}\mathcal{J} = \mathcal{I}_{\mathcal{K}}$ and $\text{Im} \mathcal{J} = \mathcal{V}$ and where $\mathcal{A} \subseteq \mathfrak{B}(\mathcal{K})$ is a $*$ -subalgebra of $\mathfrak{B}(\mathcal{K})$ for some Hilbert space $\mathcal{K}$, $\dim(\mathcal{K}) \leq \dim(\mathcal{H})$. For later convenience we shall also define the projector $\mathcal{Q} \equiv \mathcal{I} - \mathcal{P}$. Note that neither $\mathcal{P}$ or $\mathcal{Q}$ need to be orthogonal, i.e. self adjoint.

To provide some intuition on these mathematical objects we next anticipate the reduction and injection maps we are going to use in the applications of Sec.V:

- Let us assume a bipartite case, i.e. $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_B$. We can then consider $\mathcal{P}(\rho) = \text{tr}_B(\rho) \otimes \tau$ for some $\tau \in \mathfrak{D}(\mathcal{H}_B)$, $\tau > 0$ to be our CPTP projector. Then, by choosing $\mathcal{K} = \mathcal{H}_S$ and $\mathcal{A} = \mathfrak{B}(\mathcal{H}_S)$ we have the maps $\mathcal{R}(\rho) = \text{tr}_B(\rho)$ and $\mathcal{J}(\sigma) = \sigma \otimes \tau$.
- Another CPTP projector we might want to consider is the projector onto the diagonal matrix $\mathcal{P}(\rho) = \sum_{j=0}^{n-1} |j\rangle\langle j| \rho |j\rangle\langle j|$ in with case $\mathcal{K} = \mathcal{H}$, and $\mathcal{A}$ is the commutative algebra given by $\mathcal{A} \equiv \text{span}\{|j\rangle\langle j|, j = 0, \dots, n-1\} \simeq \mathbb{C}^n$. Then the maps $\mathcal{R}(\rho) = \sum_{j=0}^{n-1} |j\rangle\langle j| \rho |j\rangle\langle j|$ in which case $\mathcal{R}(\rho) \in \mathbb{C}^n$, and $\mathcal{J}(\sigma) = \sum_{j=0}^{n-1} |j\rangle\langle j| \sigma |j\rangle\langle j|$ factorize $\mathcal{P}$.

The reduction map $\mathcal{R}$ determines the reduced state – the one we want to keep track of: Let $\sigma_t \in \mathfrak{D}(\mathcal{K})$ be defined as $\sigma_t = \mathcal{R}(\rho_t)$ for all $t \geq 0$. The evolution of the reduced state is then given by

$$\sigma_t = \mathcal{R}e^{\mathcal{L}t}(\rho_0) = \mathcal{R}e^{\mathcal{L}t}\mathcal{J}(\sigma_0) + \mathcal{R}e^{\mathcal{L}t}\mathcal{Q}(\rho_0).$$

We next introduce an important simplifying assumption that will hold throughout the rest of the paper.

Assumption 1. We assume that $\mathcal{P}(\rho_0) = \rho_0$.

[Add a comment on the assumption and say that this is often assumed to be true.] This assumption directly implies that the evolution of the reduced state takes the form:

$$\sigma_t = \mathcal{R}e^{\mathcal{L}t}\mathcal{J}(\sigma_0). \quad (3)$$

In this work we are interested in characterizing a time-dependent generator that approximate well the evolution of σ_t for $t \geq 0$. To that aim let us define a density operator $\mu_t \in \mathfrak{D}(\mathcal{H})$ whose evolution is described by

$$\begin{cases} \dot{\mu}_t = \mathcal{F}_t(\mu_t) \\ \mu_0 = \sigma_0 \end{cases} \quad (4)$$

and where $\mathcal{F}_t$ is a time-dependent generator. Our aim is then to construct $\mathcal{F}_t$ so that $\mu_t \approx \sigma_t$. In other words we aim to determine $\mathcal{F}_t$ so that

$$\mathbb{T}e^{\int_0^t \mathcal{F}_s ds} \approx \mathcal{R}e^{\mathcal{L}t}\mathcal{J}$$

where $\mathbb{T}$ denotes the time-ordering operator. The sense in which the approximation $\approx$ is intended shall be defined later.

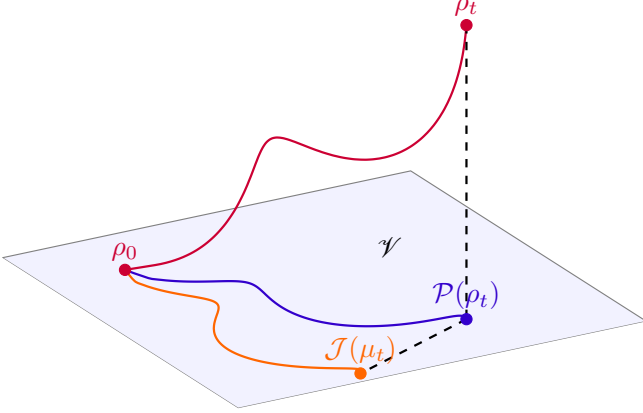


Figure 1: Graphical representation of the approximation problem. The pale blue plane represents the operator space $\mathcal{V}$ onto which $\mathcal{P}$ projects. The red curve represents the trajectory of the original model ρ_t , which starts in $\mathcal{V}$ at $t = 0$. The blue curve represents the projection of the true trajectory $\mathcal{P}(\rho_t)$, i.e. the one we want to approximate. Finally, the orange curve represents the approximate evolution we aim to find, i.e. $\mathcal{J}(\mu_t)$.

Desirable properties for $\mathcal{F}_t$ certainly include the fact that we would like $\mathbb{T}e^{\int_0^t \mathcal{F}_s ds}$ to be CPTP for all $t \geq 0$. Note that necessary conditions to impose on $\mathcal{F}_t$ so that $\mathbb{T}e^{\int_0^t \mathcal{F}_s ds}$ is CPTP for all $t \geq 0$ are not known [1].

A graphical representation of the approximation problem is given in Fig. 1.

B. Literature review

Make connections and comparisons with other works in the literature.

C. Time-convolutionless master equation

Before moving to the method proposed in this work for computing the time dependent generator $\mathcal{F}_t$ we shall discuss the derivation of time-convolutionless master equations, which is the standard method found in the literature [3–6] to solve this issue.

Dividing the evolution of ρ_t in the evolution of $\mathcal{P}\rho_t$ and that of $\mathcal{Q}\rho_t$ we have

$$\begin{aligned} \frac{d}{dt}\mathcal{P}\rho_t &= \mathcal{P}\mathcal{L}\mathcal{P}\rho_t + \mathcal{P}\mathcal{L}\mathcal{Q}\rho_t, \\ \frac{d}{dt}\mathcal{Q}\rho_t &= \mathcal{Q}\mathcal{L}\mathcal{P}\rho_t + \mathcal{Q}\mathcal{L}\mathcal{Q}\rho_t. \end{aligned}$$

Solving the second equation for $\mathcal{Q}\rho_t$ we obtain

$$\mathcal{Q}\rho_t = e^{\mathcal{Q}\mathcal{L}\mathcal{Q}t}\mathcal{Q}\rho_0 + \int_0^t e^{\mathcal{Q}\mathcal{L}\mathcal{Q}s}\mathcal{Q}\mathcal{L}\mathcal{P}\rho_{t-s}ds$$

and substituting this equation into the first equation we obtain the famous Nakajima-Zwanzig equation [7, 8]:

$$\begin{aligned} \frac{d}{dt}\mathcal{P}\rho_t &= \mathcal{P}\mathcal{L}\mathcal{P}\rho_t + \mathcal{P}\mathcal{L}e^{\mathcal{Q}\mathcal{L}\mathcal{Q}t}\mathcal{Q}\rho_0 \\ &\quad + \int_0^t \mathcal{P}\mathcal{L}\mathcal{Q}e^{\mathcal{Q}\mathcal{L}\mathcal{Q}s}\mathcal{Q}\mathcal{L}\mathcal{P}\rho_{t-s}ds. \end{aligned}$$

In order to derive the time-convolutionless master equation we recall that $\rho_{t-s} = e^{-\mathcal{L}s}\rho_t$ so that

$$\mathcal{Q}\rho_t = e^{\mathcal{Q}\mathcal{L}\mathcal{Q}t}\mathcal{Q}\rho_0 + \int_0^t e^{\mathcal{Q}\mathcal{L}\mathcal{Q}s}\mathcal{Q}\mathcal{L}\mathcal{P}e^{-\mathcal{L}s}ds \cdot (\mathcal{P} + \mathcal{Q})\rho_t.$$

Now, defining

$$\mathcal{E}_t \equiv \mathcal{I} - \int_0^t e^{\mathcal{Q}\mathcal{L}\mathcal{Q}s}\mathcal{Q}\mathcal{L}\mathcal{P}e^{-\mathcal{L}s}ds$$

we have

$$\mathcal{E}_t\mathcal{Q}\rho_t = e^{\mathcal{Q}\mathcal{L}\mathcal{Q}t}\mathcal{Q}\rho_0 + (\mathcal{I} - \mathcal{E}_t)\mathcal{P}\rho_t$$

If we then assume that $\mathcal{E}_t$ is invertible for all $t \geq 0$ we obtain

$$\mathcal{Q}\rho_t = \mathcal{E}_t^{-1}e^{\mathcal{Q}\mathcal{L}\mathcal{Q}t}\mathcal{Q}\rho_0 + (\mathcal{E}_t^{-1} - \mathcal{I})\mathcal{P}\rho_t$$

Substituting this inside the equation above we obtain

$$\begin{aligned} \frac{d}{dt}\mathcal{P}\rho_t &= \mathcal{P}\mathcal{L}\mathcal{P}\rho_t + \mathcal{P}\mathcal{L}\mathcal{E}_t^{-1}e^{\mathcal{Q}\mathcal{L}\mathcal{Q}t}\mathcal{Q}\rho_0 + \mathcal{P}\mathcal{L}(\mathcal{E}_t^{-1} - \mathcal{I})\mathcal{P}\rho_t \\ &= \mathcal{P}\mathcal{L}\mathcal{E}_t^{-1}e^{\mathcal{Q}\mathcal{L}\mathcal{Q}t}\mathcal{Q}\rho_0 + \mathcal{P}\mathcal{L}\mathcal{E}_t^{-1}\mathcal{P}\rho_t. \end{aligned}$$

Note that under the assumptions that $\mathcal{P}\rho_0 = \rho_0$ we have that $\mathcal{Q}\rho_0 = 0$ and thus the equation gets simplified to

$$\frac{d}{dt}\mathcal{P}\rho_t = \mathcal{P}\mathcal{L}\mathcal{E}_t^{-1}\mathcal{P}\rho_t.$$

[Separate the projector into $\mathcal{R}$ and $\mathcal{J}$.]
[Describe the drawbacks of this approach.]

III. POLYNOMIAL APPROXIMATIONS OF THE TIME-DEPENDENT GENERATOR

In this section we derive an approximation method that is applicable to any LTI system. For this reason, only in this section, we relax the assumptions introduced in the previous section and that will become important in the next section. Specifically, in this section, we regard ρ_t as the state of any LTI system $\mathcal{L}$ as any LTI generator and $\mathcal{P}$ as any projector. For all intended purposes, in this section we can assume $\rho_t \in \mathbb{C}^n$, $\mathcal{L}, \mathcal{P} \in \mathbb{C}^{n \times n}$. Nevertheless, our goal remains to find an approximation of the reduced generator $\mathcal{F}_t \in \mathbb{C}^{m \times m}$ where $m = \dim \text{Im} \mathcal{P}$ that govern the dynamics of the reduced state $\sigma_t \in \mathbb{C}^m$.

We next state the main result of this work.

Theorem 1. *Let us define*

$$\mathcal{F}_{t,N} \equiv \sum_{k=0}^N t^k \mathcal{F}_{(k+1)} \quad (5)$$

where

$$\mathcal{F}_{(k)} \equiv k \frac{\mathcal{R}\mathcal{L}^k \mathcal{J}}{k!} - \sum_{h=1}^{k-1} \mathcal{F}_{(k-h)} \frac{\mathcal{R}\mathcal{L}^h \mathcal{J}}{h!}. \quad (6)$$

Then:

$$\mathcal{R}e^{\mathcal{L}t} \mathcal{J} - \mathbb{T}e^{\int_0^t \mathcal{F}_{s,N} ds} = O(t^{N+1}). \quad (7)$$

Before proceeding to prove this result we shall observe a few key aspects. It is important to notice that the right-hand side of equation (6) depends on $\mathcal{R}\mathcal{L}^k \mathcal{J}$ and $\{\mathcal{F}_{(s)}\}_{s=1}^{k-1}$. This means that equation (6) can be used iteratively to construct the terms $\mathcal{F}_{(k)}$ starting from $k = 1$ and growing up to $k = N$. For the reader's convenience we write here the first few terms of the series:

$$\begin{aligned} \mathcal{F}_{(1)} &= \mathcal{R}\mathcal{L}\mathcal{J} \\ \mathcal{F}_{(2)} &= \mathcal{R}\mathcal{L}^2 \mathcal{J} - \mathcal{F}_{(1)} \mathcal{R}\mathcal{L}\mathcal{J} = \mathcal{R}\mathcal{L}^2 \mathcal{J} - (\mathcal{R}\mathcal{L}\mathcal{J})^2 \\ \mathcal{F}_{(3)} &= \frac{\mathcal{R}\mathcal{L}^3 \mathcal{J}}{2} - \mathcal{F}_{(2)} \mathcal{R}\mathcal{L}\mathcal{J} - \mathcal{F}_{(1)} \frac{\mathcal{R}\mathcal{L}^2 \mathcal{J}}{2} \\ &= \frac{\mathcal{R}\mathcal{L}^3 \mathcal{J}}{2} - \frac{\mathcal{F}_{(1)} \mathcal{F}_{(2)}}{2} - \frac{\mathcal{F}_{(1)}^3}{2} - \mathcal{F}_{(2)} \mathcal{F}_{(1)} \end{aligned}$$

It is interesting to notice a couple of points about the first few terms: i) As one would expect, the term of order one $\mathcal{F}_{(1)}$ coincides with the reduction of the original dynamics which is often used as rough time-independent approximation of the reduced evolution [CITE]; ii) The second order term does not only contain the reduction of the generator squared $\mathcal{R}\mathcal{L}^2 \mathcal{J}$ but also contains a correction term depends on the first order term $\mathcal{F}_{(1)}$.

[Add comment on the approximation we are making.]

Assuming the polynomial structure given in equation (5) for the approximated generator is clearly restrictive but this assumption greatly simplifies the computation of the time-ordered exponential $\mathbb{T}e^{\int_0^t \mathcal{F}_{s,N} ds}$ as shown in the next key result.

Theorem 2. *Let us assume that $\mathcal{F}_{t,N} = \sum_{k=0}^N t^k \mathcal{F}_{(k+1)}$. Then the time-ordered exponential can be written as $\mathbb{T}e^{\int_0^t \mathcal{F}_{s,N} ds} = \sum_{k=0}^{+\infty} t^k \mathcal{E}_{(k)}$ where $\mathcal{E}_{(0)} = \mathcal{I}$ and*

$$\mathcal{E}_{(k)} = \frac{1}{k} \sum_{s=1}^k \mathcal{F}_{(s)} \mathcal{E}_{(k-s)}$$

and where we used the convention $\mathcal{F}_{(k)} = 0$ for all $k > N$.

Proof of this result and an explanatory example are given in Appendix A 1.

Note that Theorem 2 provides equation (2), a closed formula to compute the coefficients of the time-ordered exponential $\mathcal{E}_{(k)}$ as a function of the terms $\mathcal{F}_{(k)}$. Furthermore, as in equation (2) only the terms $\{\mathcal{E}_{(\ell)}\}_{\ell=1}^{k-1}$ appear, this equation can be used iteratively to compute each term $\mathcal{E}_{(k)}$.

We are finally ready to prove Theorem 1.

Proof. First of all we can use Theorem 2 to write

$$\mathbb{T}e^{\int_0^t \mathcal{F}_{s,N} ds} = \sum_{f=0}^{+\infty} t^f \mathcal{E}_{(f)}$$

where $\mathcal{E}_{(k)} = \frac{1}{k} \sum_{s=1}^k \mathcal{F}_{(s)} \mathcal{E}_{(k-s)} = \frac{1}{k} \sum_{h=0}^{k-1} \mathcal{F}_{(k-h)} \mathcal{E}_{(h)}$ and where we reversed the order of the summation by imposing $h = k - s$. We next prove that, for $1 \leq k \leq N$, we have $\mathcal{E}_{(k)} = \frac{\mathcal{R}\mathcal{L}^k \mathcal{J}}{k!}$ by induction. The case $k = 1$ is trivial as $\mathcal{E}_{(1)} = \mathcal{F}_{(1)} \mathcal{E}_{(0)} = \mathcal{R}\mathcal{L}\mathcal{J}$. Let then assume that $\mathcal{E}_{(h)} = \frac{\mathcal{R}\mathcal{L}^h \mathcal{J}}{h!}$ for all $h < k$ and let us prove that $\mathcal{E}_{(k)} = \frac{\mathcal{R}\mathcal{L}^k \mathcal{J}}{k!}$. Then

$$\begin{aligned} \mathcal{E}_{(k)} &= \frac{1}{k} \sum_{h=0}^{k-1} \mathcal{F}_{(k-h)} \mathcal{E}_{(h)} = \frac{\mathcal{F}_{(k)}}{k} + \frac{1}{k} \sum_{h=1}^{k-1} \mathcal{F}_{(k-h)} \mathcal{E}_{(h)} \\ &= \frac{\mathcal{F}_{(k)}}{k} + \frac{1}{k} \sum_{h=1}^{k-1} \mathcal{F}_{(k-h)} \frac{\mathcal{R}\mathcal{L}^h \mathcal{J}}{h!} = \frac{\mathcal{R}\mathcal{L}^k \mathcal{J}}{k!} \end{aligned}$$

where we simply substituted the definition of $\mathcal{F}_{(k)}$ given in eq. (6).

By leveraging the definition of the exponential map $e^{\mathcal{L}t}$ we can expand the term $\mathcal{R}e^{\mathcal{L}t} \mathcal{J}$ to obtain

$$\mathcal{R}e^{\mathcal{L}t} \mathcal{J} = \sum_{k=0}^{+\infty} \frac{t^k}{k!} \mathcal{R}\mathcal{L}^k \mathcal{J}.$$

Combining the two terms we obtain

$$\begin{aligned} \mathcal{R}e^{\mathcal{L}t} \mathcal{J} - \mathbb{T}e^{\int_0^t \mathcal{F}_{s,N} ds} &= \sum_{k=0}^{+\infty} t^k \left(\frac{\mathcal{R}\mathcal{L}^k \mathcal{J}}{k!} - \mathcal{E}_{(k)} \right) \\ &= \sum_{k=0}^N t^k \left(\frac{\mathcal{R}\mathcal{L}^k \mathcal{J}}{k!} - \mathcal{E}_{(k)} \right) + O(t^{N+1}) \end{aligned}$$

which concludes the proof. $\square$

A. Testbed

To showcase the capability of the method we here consider a simple numerical example. The code necessary to reproduce the numerical experiment is available at [9]. Specifically we consider a random autonomous and stable system of size $n = 20$. Let then $\mathcal{A}$ be a random uniform matrix in $\mathbb{C}^{n \times n}$, i.e. $[\mathcal{A}]_{i,j} \sim \mathcal{U}([0, 1])$ and let $\mathcal{B} = \mathcal{A} - M \mathbb{1}_{n \times n}$ where M is the absolute value of largest

eigenvalue of $\mathcal{A}$. The dynamic's generator $\mathcal{L}$ is then obtained as $\mathcal{L} = \frac{1}{2} \frac{\mathcal{B}}{\|\mathcal{B}\|_{op}}$ where $\|\cdot\|_{op}$ denotes the operator norm. This ensures that $\mathcal{L}$ is simply stable and has operator norm $\|\mathcal{L}\|_{op} = \frac{1}{2}$.

The projector is then constructed through $\mathcal{R} = \begin{bmatrix} \mathbb{1}_{m \times m} & 0_{m \times n-m} \\ 0_{m \times n-m} & 0_{n-m \times n-m} \end{bmatrix}$ and $\mathcal{J} = \mathcal{R}^\dagger$ to obtain

$$\mathcal{P} = \begin{bmatrix} \mathbb{1}_{m \times m} & 0_{m \times n-m} \\ 0_{m \times n-m} & 0_{n-m \times n-m} \end{bmatrix}.$$

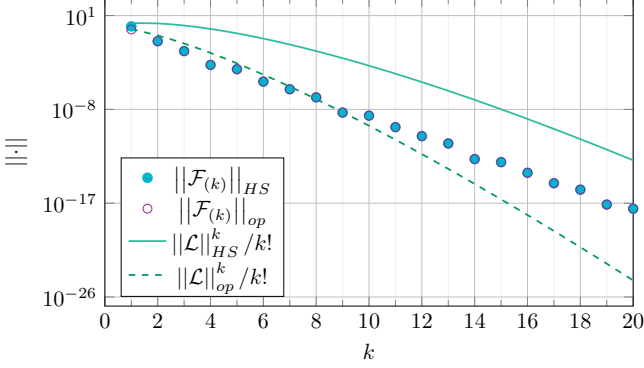


Figure 2: Operator and Hilbert-Schmidt norm of the terms $\mathcal{F}_{(k)}$.

By using eq. (6) we can then compute the terms $\mathcal{F}_{(k)}$ iteratively for $k = 1, \dots, 20$. Fig. 2 shows the both the Hilbert-Schmidt and operator norm of the different terms $\mathcal{F}_{(k)}$. Interestingly, the Hilbert-Schmidt norm, $\|\mathcal{F}_{(k)}\|_{HS}$, is bounded by $\frac{\|\mathcal{L}\|_{HS}^k}{k!}$, indicating that the series $\mathcal{F}_{t,N}$ should be converging for $N \rightarrow \infty$. **Curiously, the same does not hold for the operator norm.**

IV. COMPLETE POSITIVITY OF THE APPROXIMATE MODEL AT FIRST AND SECOND ORDER

A. First order

In this section we recall relevant known results about CPTP projectors, their images and reduced dynamics onto distorted algebras from the literature [10–16]. The consequences of the assumptions we made about the projector $\mathcal{P}$ are summarized in the following proposition.

Proposition 1. *Let $\mathcal{P}$ be a CPTP projector with a full rank fixed point $\bar{\rho} > 0$, $\mathcal{P}(\bar{\rho}) = \bar{\rho}$ and let $\mathcal{V}$ be its image, i.e. $\mathcal{V} \equiv \text{Im}\mathcal{P} \subseteq \mathfrak{B}(\mathcal{H})$. Then there exist a decomposition of the Hilbert space*

$$\mathcal{H} = \bigoplus_k (\mathcal{H}_{F,k} \otimes \mathcal{H}_{G,k}),$$

a unitary operator $U \in \mathfrak{B}(\mathcal{H})$, and a set of density oper-

ators $\tau_k \in \mathfrak{D}(\mathcal{H}_{G,k})$ such that

$$\mathcal{V} = U \left(\bigoplus_k \mathfrak{B}(\mathcal{H}_{F,k}) \otimes \tau_k \right) U^\dagger. \quad (8)$$

Furthermore $\mathcal{V}$ is a $$ -algebra closed with respect to the product $X \cdot_\sigma Y \equiv X \sigma Y$ where*

$$\sigma = U \left(\bigoplus_k \mathbb{1}_{F,k} \otimes \tau_k^{-1} \right) U^\dagger.$$

Let then W_k be a linear operator from $\mathcal{H}$ onto $\mathcal{H}_{F,k} \otimes \mathcal{H}_{G,k}$ such that $W_k W_k^\dagger = \mathbb{1}_{F,k} \otimes \mathbb{1}_{G,k}$ [11]. Then the CPTP projection $\mathcal{P}$ onto $\mathcal{V}$ takes the form

$$\mathcal{P}(X) = U \left[\bigoplus_k \text{tr}_{\mathcal{H}_{G,k}} (W_k X W_k^\dagger) \otimes \tau_k \right] U^\dagger. \quad (9)$$

and is orthogonal with respect to the inner product $\langle X, Y \rangle_{\sigma, \lambda} = \text{tr}[X^\dagger \sigma^\lambda Y \sigma^{1-\lambda}]$ for $\lambda \in [0, 1]$.

Proof of this result can be found in [10] and [11, 12, 16–18]. Note that projectors of this type take the name of state extensions and are the dual of conditional expectations [19]. Furthermore, note that a $*$ -algebra closed with respect to a modified product often takes the name of *distorted algebra* [14], while the block structure provided in Eq. (8) takes the name of *Wedderburn decomposition*.

The previous result provides us with a structure for the image of a CPTP projection whenever there exists a full rank state $\bar{\rho}$ such that $\mathcal{P}(\bar{\rho}) = \bar{\rho}$. From this structure one can observe that a distorted algebra $\mathcal{V}$ is isomorphic to a smaller representation which we denote $\mathcal{A}$ where the weighted copies are removed, i.e. $\mathcal{A} \equiv \bigoplus_k \mathfrak{B}(\mathcal{H}_{F,k})$. This allows us to find the two factors $\mathcal{R} : \mathfrak{B}(\mathcal{H}) \rightarrow \mathcal{A}$ and $\mathcal{J} : \mathcal{A} \rightarrow \mathfrak{B}(\mathcal{H})$ such that, $\mathcal{R}\mathcal{J} = \mathcal{I}_{\mathcal{A}}$ and $\mathcal{P} = \mathcal{J}\mathcal{R}$. Furthermore, one can observe [15] that these two factors can be chosen to be CPTP, i.e. for all $X \in \mathfrak{B}(\mathcal{H})$

$$\mathcal{R}(X) = \bigoplus_k \text{tr}_{\mathcal{H}_{G,k}} (W_k X W_k^\dagger) = \bigoplus_k X_{F,k} = \check{X}, \quad (10)$$

and for all $\check{X} \in \mathcal{A}$, $\check{X} = \bigoplus_k X_{F,k}$,

$$\mathcal{J}(\check{X}) = U \left(\bigoplus_k X_{F,k} \otimes \tau_k \right) U^\dagger. \quad (11)$$

The reduction and injection maps given in equations (10) and (11) then allow to compute the reduced dynamics $\mathcal{R}\mathcal{L}\mathcal{J}$ restricted onto the algebra $\mathcal{A}$. One can then prove that the reduced generator $\check{\mathcal{L}}$ is GKLS by leveraging the structure of the reduction and injection maps. We here report [15, Theorem 4] for completeness.

Theorem 3. *Let $\mathcal{P}$ be a CPTP projection with a full rank fixed point, let $\mathcal{V}$ be its image $\mathcal{V} = \text{Im}\mathcal{P} \subseteq \mathfrak{B}(\mathcal{H})$, and let $\mathcal{R}$ and $\mathcal{J}$ denote the CPTP factorization of $\mathcal{P} = \mathcal{J}\mathcal{R}$,*

as defined in Eq.(10) and (11). Then for any Lindblad generator $\mathcal{L}$, its restriction to $\mathcal{A}$, $\mathcal{RLJ}$, is also a Lindblad generator, that is, $\mathcal{RLJ} : \mathcal{A} \rightarrow \mathcal{A}$ and $\{e^{\tilde{\mathcal{L}}t}\}_{t \geq 0}$ is a quantum dynamical semi-group.

Note that this theorem only proves the fact that $\mathcal{RLJ}$ is of Lindblad type. In order to compute the reduced Hamiltonian and noise operators one can use the procedure described in [20, Appendix].

B. Second order: The case of purely Hamiltonian evolution

While the results we derived up to this point hold for general Lindblad generators $\mathcal{L}$, we shall now focus on the special case of purely Hamiltonian evolution, i.e. $\mathcal{L}(\cdot) = -i[H, \cdot]$. Other than the fact that this assumption is interesting in practical cases, we shall see that under this simplifying assumption we are able to prove that the second order term $\mathcal{F}_{(2)}$ is also of Lindblad-type.

The next result is based on the fact that the squared of a purely Hamiltonian generator is equal to a dissipator:

$$\begin{aligned} \mathcal{L}^2(\rho) &= -[H[H, \rho]] = -(H^2\rho - 2H\rho H + \rho H^2) \\ &= 2H\rho H - \{H^2, \rho\} = \mathcal{D}_{\sqrt{2}H}(\rho). \end{aligned} \quad (12)$$

This fact allows us to prove the next result.

Theorem 4. *Under the assumptions of Theorem 3 we have that $\mathcal{F}_{(2)} = \mathcal{RL}^2\mathcal{J} - (\mathcal{RLJ})^2$ is of Lindblad-type and for $\mathcal{F}_{t,2} \equiv \mathcal{F}_{(1)} + t\mathcal{F}_{(2)}$ we have that $\mathbb{T}e^{\int_0^t \mathcal{F}_{s,2} ds}$ forms a set of CPTP maps for all $t \geq 0$.*

Proof. By leveraging [15, Proposition 5] we have that $\mathcal{RLJ}(\tilde{\rho}) = -i[\tilde{H}, \tilde{\rho}]$ where $\tilde{H} = \mathcal{J}^\dagger(H)$ hence, using equation (12) we obtain $(\mathcal{RLJ})^2(\tilde{\rho}) = \mathcal{D}_{\sqrt{2}\tilde{H}}(\tilde{\rho})$. On the other hand, by leveraging [20, Proposition 6 and Proposition 7], we have that $\mathcal{RD}_{\sqrt{2}H}\mathcal{J}(\tilde{\rho}) = \sum_{\tilde{C} \in \mathcal{X}(\sqrt{2}H)} \mathcal{D}_{\tilde{C}}(\tilde{\rho})$ for a set of operators $\mathcal{X}(\sqrt{2}H) \subset \mathfrak{B}(\mathcal{K})$. Importantly, by [20, Corollary 4], we have that $\mathcal{J}^\dagger(H) \in \mathcal{X}(\sqrt{2}H)$ and thus we can write $\mathcal{RD}_{\sqrt{2}H}\mathcal{J}(\tilde{\rho}) = \sum_{\tilde{C} \in \mathcal{X}(\sqrt{2}H)/\{\sqrt{2}\tilde{H}\}} \mathcal{D}_{\tilde{C}}(\tilde{\rho}) + \mathcal{D}_{\sqrt{2}\tilde{H}}(\tilde{\rho})$ since $\sqrt{2}\tilde{H} = \mathcal{J}^\dagger(\sqrt{2}H)$. Then, by combining these two results we obtain

$$\mathcal{F}_{(2)} = \mathcal{RL}^2\mathcal{J} - (\mathcal{RLJ})^2 = \sum_{\tilde{C} \in \mathcal{X}(\sqrt{2}H)/\{\sqrt{2}\tilde{H}\}} \mathcal{D}_{\tilde{C}}(\tilde{\rho})$$

which is the sum of a set of GKLS, purely dissipative generators and thus of Lindblad type. Combining this with Theorem 3 it trivially follows that $\mathcal{F}_{t,2}$ is of Lindblad type for all $t \geq 0$. $\square$

C. Second order – Special case: Bipartite systems

Because the proof of Theorem 4 is based on a few very technical results, we here want to present an equivalent pedagogical derivation of the same result. Note that this derivation is quite relevant in practice as it focuses on the most studied case of a bi-partite system but the cases included in the assumptions of Theorem 4 are more general. This exact case will be the focus of the two applications of interest studied in Sec V A and V B.

Let us assume that $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_E$ and let assume that the CPTP projector $\mathcal{P}$ is given by the $\mathcal{P}(\cdot) = \text{tr}_E(\cdot) \otimes \tau$ for some full rank state $\tau \in \mathfrak{D}(\mathcal{H}_E)$. The projector $\mathcal{P}$ is then factorized into the reduction map $\mathcal{R}(\cdot) = \text{tr}_E(\cdot)$ and the injection map $\mathcal{J}(\cdot) = \cdot \otimes \tau$. In line with the assumptions of this section we assume that $\mathcal{L}(\cdot) = -i\text{ad}_H(\cdot)$. Given the bipartite structure of the Hilbert space $\mathcal{H}$ we can always write $H = \sum_k S_k \otimes E_k$ for two sets of Hermitian operators $\{S_k\} \subset \mathfrak{B}(\mathcal{H}_S)$ and $\{E_k\} \subset \mathfrak{B}(\mathcal{H}_E)$.

We can then proceed to compute the reduced generator approximated at second order $\mathcal{F}_{t,2}(\mu)$. First, using [15, Proposition 5], we have that the generator at first order is

$$\mathcal{F}_{(1)}(\mu) = \mathcal{RLJ}(\mu) = -i[\tilde{H}, \mu]$$

where

$$\begin{aligned} \tilde{H} &= \mathcal{J}^\dagger(H) = \text{tr}_E[(\mathbb{1}_S \otimes \tau)H] = \sum_k \text{tr}_E[S_k \otimes (\tau E_k)] \\ &= \sum_k a_k(\tau) S_k \end{aligned}$$

and $a_k(\tau) \equiv \text{tr}[\tau E_k]$. Next, we can proceed to compute $\mathcal{F}_{(2)}$ by first computing $\mathcal{RL}^2\mathcal{J}$:

$$\begin{aligned}
\mathcal{RL}^2 \mathcal{J}(\mu) &= \sum_{j,k} \text{tr}_E [2(S_j \otimes E_j)(\mu \otimes \tau)(S_k \otimes E_k) - (S_k \otimes E_k)(S_j \otimes E_j)(\mu \otimes \tau) - (\mu \otimes \tau)(S_k \otimes E_k)(S_j \otimes E_j)] \\
&= \sum_{j,k} 2S_j \mu S_k \underbrace{\text{tr}[E_j \tau E_k]}_{\equiv c_{j,k}(\tau)} - S_k S_j \mu \underbrace{\text{tr}[E_k E_j \tau]}_{=c_{j,k}(\tau)} - \mu S_k S_j \underbrace{\text{tr}[\tau E_k E_j]}_{=c_{j,k}(\tau)} \\
&= \sum_{j,k} c_{j,k}(\tau) [2S_j \mu S_k - \{S_k S_j, \mu\}].
\end{aligned}$$

On the other hand, since $(\mathcal{RL}\mathcal{J})^2 = \mathcal{D}_{\sqrt{2}\tilde{H}}$, we have

$$(\mathcal{RL}\mathcal{J})^2(\eta) = \sum_{j,k} a_j(\tau) a_k(\tau) [2S_j \eta S_k - \{S_k S_j, \eta\}]$$

and thus

$$\begin{aligned}
\mathcal{F}_{(2)}(\mu) &= [\mathcal{RL}^2 \mathcal{J} - (\mathcal{RL}\mathcal{J})^2](\mu) = \\
&= \sum_{j,k} [\chi(\tau)]_{j,k} [2S_j \mu S_k - \{S_k S_j, \mu\}]
\end{aligned}$$

where we defined the matrix $\chi(\tau)$ whose elements are $[\chi(\tau)]_{j,k} \equiv c_{j,k}(\tau) - a_j(\tau) a_k(\tau)$. One can then notice that the matrix $\chi(\tau)$ coincides with a covariance matrix, i.e.

$$\begin{aligned}
[\chi(\tau)]_{j,k} &= \text{tr}[E_j \tau E_k] - \text{tr}[E_j \tau] \text{tr}[E_k \tau] \\
&= \mathbb{E}_\tau[(E_k - \mathbb{E}_\tau[E_k] \mathbb{1}_E)(E_j - \mathbb{E}_\tau[E_j] \mathbb{1}_E)]
\end{aligned}$$

hence $\chi(\tau)$ is a positive semi-definite matrix. Since $\chi(\tau)$ is positive semi-definite, it is diagonalizable with a unitary matrix, i.e. $UU^\dagger = U^\dagger U = \mathbb{1}$ such that $U\chi(\tau)U^\dagger = \gamma(\tau)$ where $\gamma(\tau)$ is a diagonal matrix of positive values (note that in reality U is also a function of τ). We can then define

$$L_h = \sum_j \overline{[U]_{h,j}} S_j$$

to obtain

$$\begin{aligned}
\mathcal{F}_{(2)}(\mu) &= \sum_{j,k} [\chi(\tau)]_{j,k} [2S_j \mu S_k - \{S_k S_j, \mu\}] \\
&= \sum_{j,k,h} \underbrace{[U^\dagger]_{j,h} [\gamma(\tau)]_h [U]_{h,k}}_{[\chi(\tau)]_{j,k}} [2S_j \mu S_k - \{S_k S_j, \mu\}] \\
&= \sum_{j,k,h} [\gamma(\tau)]_h \overline{[U]_{h,j}} [U]_{h,k} [2S_j \mu S_k - \{S_k S_j, \mu\}] \\
&= \sum_h [\gamma(\tau)]_h [2L_h \mu L_h^\dagger - \{L_h^\dagger L_h, \mu\}] \\
&= \sum_h [\gamma(\tau)]_h \mathcal{D}_{\sqrt{2}L_h}(\mu).
\end{aligned}$$

Because $\chi(\tau)$ is positive-semidefinite and thus $[\gamma(\tau)]_h$ are non-negative, then $\mathcal{F}_{(2)}$ is of Lindblad type. This implies

that the approximated time-dependent generator approximated at second order

$$\mathcal{F}_{t,2}(\mu) = -i[\tilde{H}, \mu] + \sum_h t[\gamma(\tau)]_h \mathcal{D}_{\sqrt{2}L_h}(\mu)$$

is also of Lindblad type at all times $t \geq 0$ as proved in Theorem 4.

Note that in deriving the reduced generator $\mathcal{F}_{t,2}$ there was no need to move the system's description into rotating frame. This provides another benefit of the proposed approach.

V. APPLICATIONS OF INTEREST

A. Dephasing spin boson model

This section is intended as a comparison with a well established model in the literature. We will not be able to do better than this model but we should be able to highlight key aspects. This section is extremely difficult and important because it provides a connection for physicists.

Inspired by [21], we here consider a spin-boson model. Assume $\mathcal{H} = \mathcal{H}_S \otimes \mathcal{H}_B$ and an Hamiltonian $H_{SB}(t)$ in interaction picture given by

$$H_{SB}(t) = \sigma_z \otimes \underbrace{\left[\sum_k \lambda_k e^{-i\omega_k t} b_k + \bar{\lambda}_k e^{i\omega_k t} b_k^\dagger \right]}_{\equiv H_{B,z}(t)}$$

where b_k are annihilation operators (for mode k) and satisfy $[b_k, b_l] = \delta_{k,l} \mathbb{1}$. Assuming that $\rho_B(0) = e^{-\beta H_B}$ with $H_B = \sum_k \omega_k (b_k^\dagger b_k + \frac{1}{2})$, this model has an exact computable solutions that, according to [21, eq. (336) and (374)] is:

$$\rho_S(t) = \begin{bmatrix} [\rho_S(0)]_{0,0} & e^{-2\gamma(t)t} [\rho_S(0)]_{0,1} \\ e^{-2\gamma(t)t} [\rho_S(0)]_{1,0} & [\rho_S(0)]_{1,1} \end{bmatrix}$$

with

$$\gamma(t) = \sum_k 2|\lambda_k|^2 \coth\left(\frac{\beta\omega_k}{2}\right) \frac{1 - \cos(\omega_k t)}{\omega_k^2 t}.$$

Taking the derivative we get:

$$[\dot{\rho}_S(t)]_{0,1} = \underbrace{-2[\gamma(t) + \dot{\gamma}(t)t]}_{\equiv \xi(t)} [\rho_S(t)]_{0,1}$$

where

$$\xi(t) = -\sum_k 4|\lambda_k|^2 \coth\left(\frac{\beta\omega_k}{2}\right) \frac{\omega_k \sin(\omega_k t)}{\omega_k^2 t}$$

which corresponds to the Lindblad generator

$$\dot{\rho}_S(t) = \xi(t) \mathcal{D}_{\sigma_z}[\rho_S(t)].$$

Is this correct?

Defining $\langle X \rangle_B = \text{tr}[\rho_B(0)X]$, we have, from [21, eq.

(331)]:

$$\begin{aligned} \langle b_k^\dagger \rangle_B &= \langle b_k \rangle_B = \langle b_k^\dagger b_l^\dagger \rangle_B = \langle b_k b_l \rangle_B = 0 \\ \langle b_j^\dagger b_k \rangle_B &= \frac{\delta_{j,k}}{e^{\beta\omega_k} - 1} \end{aligned}$$

Then, using our approach, and considering $\tau = \rho_B(0)$, we have

$$\langle H_{B,z}(t) \rangle_B = \sum_k \lambda_k e^{-i\omega_k t} \langle b_k \rangle_B + \bar{\lambda}_k e^{i\omega_k t} \langle b_k^\dagger \rangle_B = 0,$$

hence

$$\tilde{H} = \sigma_z \langle H_{B,z}(t) \rangle_B = 0.$$

While, for the second order we obtain

$$\mathcal{L}^{(2)}(\rho_S) = \Delta \mathcal{D}_{\sigma_z}(\rho_S) \cdot \delta t$$

with

$$\begin{aligned} \Delta &\equiv \langle H_{B,z}(t) H_{B,z}(t) \rangle_B = \\ &= \sum_{j,k} \lambda_j \lambda_k e^{-i(\omega_j + \omega_k)t} \langle b_j b_k \rangle_B + \lambda_j \bar{\lambda}_k e^{-i(\omega_j - \omega_k)t} \langle b_j b_k^\dagger \rangle_B + \bar{\lambda}_j \lambda_k e^{-i(-\omega_j + \omega_k)t} \langle b_j^\dagger b_k \rangle_B + \bar{\lambda}_j \bar{\lambda}_k e^{-i(-\omega_j - \omega_k)t} \langle b_j^\dagger b_k^\dagger \rangle_B \\ &= \sum_j \lambda_j \bar{\lambda}_j \frac{e^{-i(\omega_j - \omega_j)t}}{e^{\beta\omega_j} - 1} + \bar{\lambda}_j \lambda_j \frac{e^{-i(-\omega_j + \omega_j)t}}{e^{\beta\omega_j} - 1} = \sum_k 2 \frac{|\lambda_k|^2}{e^{\beta\omega_k} - 1}. \end{aligned}$$

This turns out to be a very poor approximation of the generator. This could well be because we are trying to find a time-independent approximation of a time-dependent generator. In fact, if we change the second order approximation to the time dependent one:

$$\mathcal{L}_t^{(2)}(\rho_S) = \Delta \mathcal{D}_{\sigma_z}(\rho_S) \cdot t,$$

we obtain a much better approximation.

B. Central spin model

We want to showcase various aspects here: 1) the reduced model works; 2) the second order is CPTP 3) numerically we can easily go to higher orders 4) it works even with dissipation.

C. Ising spin chain

Here we want to show that the reduced model can be computed even for cases that are not bi-partite

VI. CONCLUSIONS AND OUTLOOK

Pros of the method: 1) The method is iterative and can thus be easily tested (implemented) for small system's sizes. 2) Does not require moving to a rotating frame. 3) The method does not only work for Hamiltonian methods but also includes Lindbladian evolutions. 4) The approach is very general and does not include the

Cons of the method: 1) The method only works for short-time approximations.

Outlook:

- convergence of the considered series (lifting part Assumption 2 to get from polynomial to analytic). Say it can be interesting mathematically but practically one would not want to do it. Also explain the difficulties we could encounter (proof of thm1).
- What happens if we remove the assumption $\mathcal{P}(\rho_0) = \rho_0$

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Appendix A: Technical results and proofs

1. Time-ordered expansion

Before proving the main result, Theorem 2, we shall develop some intuition about how we prove it.

Example 1. To provide some intuition about Theorem 2 let us consider a simple case: $\mathcal{F}_t = \mathcal{F}_{(1)} + t\mathcal{F}_{(2)} + t^2\mathcal{F}_{(3)}$. Let us first compute the first few terms of the Dyson expansion [22] of $\mathbb{T}e^{\int_0^t \mathcal{F}_s ds}$, i.e. $\mathbb{T}e^{\int_0^t \mathcal{F}_s ds} = \sum_{n=0}^{+\infty} \Phi_{(n),t}$. At

zeroth order we have $\Phi_{(0),t} = \mathcal{I}$ while at orders 1, 2 and 3 we have:

$$\begin{aligned}
\Phi_{(1),t} &= \int_0^t \mathcal{F}_s ds = \int_0^t (\mathcal{F}_{(1)} + s\mathcal{F}_{(2)} + s^2\mathcal{F}_{(3)}) ds = t\mathcal{F}_{(1)} + t^2 \frac{\mathcal{F}_{(2)}}{2} + t^3 \frac{\mathcal{F}_{(3)}}{3}, \\
\Phi_{(2),t} &= \int_0^t dt_1 \int_0^{t_1} dt_2 \mathcal{F}_{t_1} \mathcal{F}_{t_2} = \sum_{j=0}^1 \int_0^t dt_1 \mathcal{F}_{t_1} \mathcal{F}_{(j+1)} \int_0^{t_1} dt_2 t_2^j = \sum_{j=0}^1 \int_0^t dt_1 \mathcal{F}_{t_1} \mathcal{F}_{(j+1)} \frac{t_1^{j+1}}{j+1} \\
&= \sum_{k,j=0}^1 \mathcal{F}_{(k+1)} \mathcal{F}_{(j+1)} \int_0^t dt_1 t_1^k \frac{t_1^{j+1}}{j+1} = \sum_{k,j=0}^1 \mathcal{F}_{(k+1)} \mathcal{F}_{(j+1)} \frac{t^{k+j+2}}{(k+j+2)(j+1)} = \sum_{k,j=1}^2 \mathcal{F}_{(k)} \mathcal{F}_{(j)} \frac{t^{k+j}}{(j+k)j} \\
&= \underbrace{\mathcal{F}_{(1)}^2 \frac{t^2}{2}}_{j=k=1} + \underbrace{\mathcal{F}_{(2)} \mathcal{F}_{(1)} \frac{t^3}{3}}_{j=1,k=2} + \underbrace{\mathcal{F}_{(1)} \mathcal{F}_{(2)} \frac{t^3}{6}}_{j=2,k=1} + O(t^4), \\
\Phi_{(3),t} &= \int_0^t dt_1 \int_0^{t_1} dt_2 \int_0^{t_2} dt_3 \mathcal{F}_{t_1} \mathcal{F}_{t_2} \mathcal{F}_{t_3} = \int_0^t dt_1 \mathcal{F}_{t_1} \underbrace{\int_0^{t_1} dt_2 \int_0^{t_2} dt_3 \mathcal{F}_{t_2} \mathcal{F}_{t_3}}_{=\sum_{j,k=1}^2 \mathcal{F}_{(k)} \mathcal{F}_{(j)} \frac{t_1^{k+j}}{(j+k)j}} \\
&= \sum_{h,k,j=1}^2 \mathcal{F}_{(h)} \mathcal{F}_{(k)} \mathcal{F}_{(j)} \int_0^t dt_1 t_1^h \frac{t_1^{k+j}}{(j+k)j} = \sum_{h,k,j=1}^2 \mathcal{F}_{(h)} \mathcal{F}_{(k)} \mathcal{F}_{(j)} \frac{t^{h+k+j}}{(h+k+j)(j+k)j} = \underbrace{\mathcal{F}_{(1)}^3 \frac{t^3}{3!}}_{h=k=j=1} + O(t^4).
\end{aligned}$$

Grouping together the terms of $\mathbb{T}e^{\int_0^t \mathcal{F}_s ds}$ by the order of the exponent of t , we have

$$\mathbb{T}e^{\int_0^t \mathcal{F}_s ds} = \underbrace{\mathcal{I}}_{\mathcal{E}_{(0)}} + \underbrace{t \mathcal{F}_{(1)}}_{\mathcal{E}_{(1)}} + \underbrace{t^2 \left(\frac{\mathcal{F}_{(2)}}{2} + \frac{\mathcal{F}_{(1)}^2}{2} \right)}_{\mathcal{E}_{(2)}} + \underbrace{t^3 \left(\frac{\mathcal{F}_{(3)}}{3} + \frac{\mathcal{F}_{(1)} \mathcal{F}_{(2)}}{6} + \frac{\mathcal{F}_{(2)} \mathcal{F}_{(1)}}{3} + \frac{\mathcal{F}_{(1)}^3}{3!} \right)}_{\mathcal{E}_{(3)}} + O(t^4)$$

which coincides with the first few terms obtained using the equation given in Theorem 2:

$$\begin{aligned}
\mathcal{E}_{(0)} &= \mathcal{I}, \\
\mathcal{E}_{(1)} &= \mathcal{F}_{(1)} \mathcal{E}_{(0)} = \mathcal{F}_{(1)}, \\
\mathcal{E}_{(2)} &= \frac{1}{2} (\mathcal{F}_{(1)} \mathcal{E}_{(1)} + \mathcal{F}_{(2)} \mathcal{E}_{(0)}) = \frac{1}{2} (\mathcal{F}_{(1)}^2 + \mathcal{F}_{(2)}), \\
\mathcal{E}_{(2)} &= \frac{1}{3} (\mathcal{F}_{(1)} \mathcal{E}_{(2)} + \mathcal{F}_{(2)} \mathcal{E}_{(1)} + \mathcal{F}_{(3)} \mathcal{E}_{(0)}) = \frac{1}{3} \left(\frac{1}{2} \mathcal{F}_{(1)}^3 + \frac{1}{2} \mathcal{F}_{(1)} \mathcal{F}_{(2)} + \mathcal{F}_{(2)} \mathcal{F}_{(1)} + \mathcal{F}_{(3)} \right).
\end{aligned}$$

Proof (Theorem 2). We shall start by proving that $\mathbb{T}e^{\int_0^t \mathcal{F}_s ds} = \sum_{n=0}^{+\infty} \Phi_{(n),t}$ where $\Phi_{(0),t} = \mathcal{I}$ and, for $n > 0$,

$$\Phi_{(n),t} = \sum_{k_1, \dots, k_n=1}^{N+1} t^{\sum_{j=1}^n k_j} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h}.$$

First of all, using the Dyson series expansion for the time-ordered exponential $\mathbb{T}e^{\int_0^t \mathcal{F}_s ds}$, see e.g. [22–24], we have that:

$$\mathbb{T}e^{\int_0^t \mathcal{F}_s ds} = \sum_{n=0}^{+\infty} \Phi_{(n),t} \quad \text{with} \quad \Phi_{(n),t} = \int_0^t dt_1 \int_0^{t_1} dt_2 \dots \int_0^{t_{n-1}} dt_n \mathcal{F}_{t_1} \mathcal{F}_{t_2} \dots \mathcal{F}_{t_n}.$$

Next, by noticing that $\Phi_{(n),t} = \int_0^t ds \mathcal{F}_s \Phi_{(n-1),s}$ we can prove the statement by induction. The case $n = 1$ is proven

by simple calculations:

$$\Phi_{(1),t} = \int_0^t ds \mathcal{F}_s = \int_0^t ds \sum_{k_1=0}^N s^{k_1} \mathcal{F}_{(k_1+1)} = \sum_{k_1=0}^N \mathcal{F}_{(k_1+1)} \int_0^t ds s^{k_1} = \sum_{k_1=0}^N \mathcal{F}_{(k_1+1)} \frac{t^{k_1+1}}{k_1+1} = \sum_{k_1=1}^{N+1} \mathcal{F}_{(k_1)} \frac{t^{k_1}}{k_1}.$$

We next assume that the statement holds for n , i.e.

$$\Phi_{(n),t} = \sum_{k_1, \dots, k_n=1}^{N+1} t^{\sum_{j=1}^n k_j} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h}.$$

Then

$$\begin{aligned} \Phi_{(n+1),t} &= \int_0^t ds \mathcal{F}_s \Phi_{(n),s} = \int_0^t ds \mathcal{F}_s \sum_{k_1, \dots, k_n=1}^{N+1} s^{\sum_{j=1}^n k_j} \frac{\mathcal{F}_{(k_n)} \dots \mathcal{F}_{(k_1)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h} \\ &= \int_0^t ds \sum_{k=0}^N s^k \mathcal{F}_{(k+1)} \sum_{k_1, \dots, k_n=1}^{N+1} s^{\sum_{j=1}^n k_j} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h} \\ &= \sum_{k=0}^N \sum_{k_1, \dots, k_n=1}^{N+1} \mathcal{F}_{(k+1)} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h} \int_0^t s^{k+\sum_{j=1}^n k_j} ds \\ &= \sum_{k=0}^N \sum_{k_1, \dots, k_n=1}^{N+1} \frac{t^{k+1+\sum_{j=1}^n k_j}}{k+1+\sum_{j=1}^n k_j} \mathcal{F}_{(k+1)} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h} \\ &= \sum_{k=1}^{N+1} \sum_{k_1, \dots, k_n=1}^{N+1} \frac{t^{k+\sum_{j=1}^n k_j}}{k+\sum_{j=1}^n k_j} \mathcal{F}_{(k)} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h} \\ &= \sum_{k_1=1}^{N+1} \sum_{k_2, \dots, k_{n+1}=1}^{N+1} \frac{t^{k_1+\sum_{j=2}^{n+1} k_j}}{k_1+\sum_{j=2}^{n+1} k_j} \frac{\mathcal{F}_{(k_1)} \mathcal{F}_{(k_2)} \dots \mathcal{F}_{(k_{n+1})}}{\prod_{j=1}^n \sum_{h=n-j+2}^{n+1} k_h} \\ &= \sum_{k_1, \dots, k_{n+1}=1}^{N+1} \frac{t^{\sum_{j=1}^{n+1} k_j}}{\sum_{j=1}^{n+1} k_j} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_{n+1})}}{\prod_{j=1}^n \sum_{h=n-j+2}^{n+1} k_h} \\ &= \sum_{k_1, \dots, k_{n+1}=1}^{N+1} t^{\sum_{j=1}^{n+1} k_j} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_{n+1})}}{\prod_{j=1}^{n+1} \sum_{h=n-j+2}^{n+1} k_h}. \end{aligned}$$

thus proving the statement.

We next prove that $\mathbb{T}e^{\int_0^t \mathcal{F}_s ds} = \sum_{k=0}^{+\infty} t^k \mathcal{E}_{(k)}$ where $\mathcal{E}_{(0)} = \mathcal{I}$ and, for $k > 0$,

$$\mathcal{E}_{(k)} = \sum_{n=1}^k \sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j=k}}^k \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h}.$$

To prove this claim we can start by noticing that each term $\Phi_{(n),t}$ contains powers t^k of order $k \geq n$. Hence, to construct $\mathcal{E}_{(k)}$ of order k we need to collect and sum all the coefficients that multiply t^k among the terms $\Phi_{(n),t}$ for $1 \leq n \leq k$. Given $\Phi_{(n),t}$ with $n \leq k$ we have that the coefficient multiplying t^k must satisfy the constraint $\sum_{j=1}^n k_j = k$. Thus, the coefficient of $\Phi_{(n),t}$ that multiply t^k are

$$\sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j=k}}^{N+1} \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h} = \sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j=k}}^k \frac{\mathcal{F}_{(k_1)} \dots \mathcal{F}_{(k_n)}}{\prod_{j=1}^n \sum_{h=n-j+1}^n k_h}$$

where we used the fact that, if any $k_j > k$ then $\sum_{j=1}^n k_j > k$. Noticing that $\prod_{j=1}^n \sum_{h=n-j+1}^n k_h = k(\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)$ and summing over $1 \leq n \leq k$ we obtain, for $k > 0$, the second claim, i.e.

$$\mathcal{E}_{(k)} = \sum_{n=1}^k \sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j=k}}^k \frac{\mathcal{F}_{(k_1)} \cdots \mathcal{F}_{(k_n)}}{k(\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)}.$$

We are now finally ready to prove the main claim of the theorem, that is: $\mathcal{E}_{(k)} = \frac{1}{k} \sum_{s=1}^k \mathcal{F}_{(s)} \mathcal{E}_{(k-s)}$. Recalling that $\mathcal{E}_{(0)} = \mathcal{I}$ we can write:

$$\begin{aligned} \frac{1}{k} \sum_{s=1}^k \mathcal{F}_{(s)} \mathcal{E}_{(k-s)} &= \sum_{s=1}^{k-1} \frac{\mathcal{F}_{(s)}}{k} \mathcal{E}_{(k-s)} + \frac{\mathcal{F}_{(k)}}{k} \\ &= \sum_{s=1}^{k-1} \sum_{n=1}^{k-s} \sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j=k-s}}^k \frac{\mathcal{F}_{(s)}}{k} \frac{\mathcal{F}_{(k_1)} \cdots \mathcal{F}_{(k_n)}}{(k-s)(\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)} + \frac{\mathcal{F}_{(k)}}{k} \\ &= \sum_{s=1}^{k-1} \sum_{n=1}^{k-1} \sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j=k-s}}^k \frac{\mathcal{F}_{(s)}}{k} \frac{\mathcal{F}_{(k_1)} \cdots \mathcal{F}_{(k_n)}}{(k-s)(\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)} + \frac{\mathcal{F}_{(k)}}{k} \end{aligned}$$

where we used the fact that $\sum_{n=k-s+1}^{k-1} \sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j=k-s}}^k \frac{\mathcal{F}_{(s)}}{k} \frac{\mathcal{F}_{(k_1)} \cdots \mathcal{F}_{(k_n)}}{(k-s)(\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)} = 0$ since for $n > k-s$ we have that $\sum_{j=1}^n k_j > k-s$ for any set of k_j . This implies that

$$\begin{aligned} \frac{1}{k} \sum_{s=1}^k \mathcal{F}_{(s)} \mathcal{E}_{(k-s)} &= \sum_{n=1}^{k-1} \sum_{s=1}^{k-1} \sum_{\substack{k_1, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j=k-s}}^k \frac{\mathcal{F}_{(s)}}{k} \frac{\mathcal{F}_{(k_1)} \cdots \mathcal{F}_{(k_n)}}{(k-s)(\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)} + \frac{\mathcal{F}_{(k)}}{k} \\ &= \sum_{n=1}^{k-1} \sum_{k_1=1}^{k-1} \sum_{\substack{k_2, \dots, k_{n+1}=1 \\ \text{s.t. } \sum_{j=2}^{n+1} k_j=k-k_1}}^k \frac{\mathcal{F}_{(k_1)}}{k} \frac{\mathcal{F}_{(k_2)} \cdots \mathcal{F}_{(k_{n+1})}}{\underbrace{(k-k_1)(\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_{h+1})}_{\sum_{h=2}^{n+1} k_h}} + \frac{\mathcal{F}_{(k)}}{k} \\ &= \sum_{n=1}^{k-1} \sum_{k_1=1}^{k-1} \sum_{\substack{k_2, \dots, k_{n+1}=1 \\ \text{s.t. } \sum_{j=1}^{n+1} k_j=k}}^k \frac{\mathcal{F}_{(k_1)} \mathcal{F}_{(k_2)} \cdots \mathcal{F}_{(k_{n+1})}}{k(\prod_{j=1}^{n+1} \sum_{h=n-j+2}^n k_h)} + \frac{\mathcal{F}_{(k)}}{k} \\ &= \sum_{n=2}^k \sum_{k_1=1}^{k-1} \sum_{\substack{k_2, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j=k}}^k \frac{\mathcal{F}_{(k_1)} \mathcal{F}_{(k_2)} \cdots \mathcal{F}_{(k_n)}}{k(\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)} + \frac{\mathcal{F}_{(k)}}{k} \\ &= \sum_{n=1}^k \sum_{k_1=1}^k \sum_{\substack{k_2, \dots, k_n=1 \\ \text{s.t. } \sum_{j=1}^n k_j=k}}^k \frac{\mathcal{F}_{(k_1)} \mathcal{F}_{(k_2)} \cdots \mathcal{F}_{(k_n)}}{k(\prod_{j=1}^{n-1} \sum_{h=n-j+1}^n k_h)} = \mathcal{E}_{(k)}, \end{aligned}$$

thus concluding the proof. $\square$

Lemma 1. Let $\|\cdot\|$ denote any sub-multiplicative norm and let us assume that there exists a constant $\ell > 0$ such that $\|\mathcal{F}_{(s)}\| < \ell^s$ for all $s \leq k$. Then for $k > 1$, $\|\mathcal{E}_{(k)}\| \leq \ell^k$.

Proof. We prove this lemma by induction. The base case, $k = 1$ comes naturally: $\|\mathcal{E}_{(1)}\| = \|\mathcal{F}_{(1)}\| \leq \ell$. We then

assume that $||\mathcal{E}_{(s)}|| \leq \ell^s$ for all $s < k$ and prove that $||\mathcal{E}_{(k)}|| \leq \ell^k$:

$$||\mathcal{E}_{(k)}|| = \left\| \frac{\mathcal{F}_{(k)}}{k} + \frac{1}{k} \sum_{s=1}^{k-1} \mathcal{F}_{(s)} \mathcal{E}_{(k-s)} \right\| \leq \frac{\ell^k}{k} + \frac{1}{k} \sum_{s=1}^{k-1} \ell^s \ell^{k-s} = \frac{\ell^k(1+k-1)}{k} = \ell^k.$$

□

I think that this shows convergence of the series $\sum_{k=0}^{+\infty} t^k \mathcal{E}_{(k)}$ for $|t\ell| < 1$ as $\left\| \sum_{k=0}^{+\infty} t^k \mathcal{E}_{(k)} \right\| = \frac{1}{1-t\ell}$.